WORKED SOLUTION

STEP II 2025, Question 2

Pure • Complex Equations with Modulus

STEP II 2025, Question 2

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

The key idea

Write $z = x + \mathrm{i}y$ and split the equation into real and imaginary parts. A modulus is always real, so it contributes nothing to the imaginary part, and the imaginary part of z^2 (or z^3) then factorises. That factorisation tells us which lines in the Argand diagram the roots must lie on, and on each line the problem collapses to a real equation in one variable.

Solution

Part (i)

(a) Let $z = x + \mathrm{i}y$ with x, y real. Then $z^2 = x^2 - y^2 + 2xy\,\mathrm{i}$, while $|z + b|$ is a real number, and so is a. Comparing imaginary parts of $z^2 + |z + b| = a$ gives

$$2xy = 0,$$

so $x = 0$ (and z is purely imaginary) or $y = 0$ (and z is purely real). This is what we had to show.

(b) Suppose $z = \mathrm{i}y$ with y real. Then $z^2 = -y^2$ and $|z + \frac{5}{2}| = \sqrt{y^2 + \frac{25}{4}}$, so we would need

$$\sqrt{y^2 + \tfrac{25}{4}} = y^2 + \tfrac{7}{2}.$$

Both sides are positive, so this is equivalent to the squared equation

$$y^2 + \tfrac{25}{4} = y^4 + 7y^2 + \tfrac{49}{4} \quad \Longleftrightarrow \quad y^4 + 6y^2 + 6 = 0.$$

The left-hand side is at least 6 for every real y, so there is no solution: the equation has no purely imaginary roots.

(c) Suppose $z = x$ is real, so we need $x^2 + |x + \frac{7}{2}| = \frac{5}{2}$. We split according to the sign of $x + \frac{7}{2}$.

- If $x \geq -\frac{7}{2}$: $x^2 + x + \frac{7}{2} = \frac{5}{2}$, that is $x^2 + x + 1 = 0$, whose discriminant is $1 - 4 = -3 < 0$. No real solutions.
- If $x < -\frac{7}{2}$: $x^2 - x - \frac{7}{2} = \frac{5}{2}$, that is $x^2 - x - 6 = (x-3)(x+2) = 0$. The roots 3 and -2 both fail the condition $x < -\frac{7}{2}$, so neither is a solution.

Hence there are no purely real roots.

Ask yourself Why is it not enough to solve $x^2 - x - 6 = 0$ and stop?

Each case only describes the original equation on its own range of x. A root of the case equation that lies outside that range solves a different equation (one with the wrong sign inside the modulus), so it must be thrown away.

(d) Now $\frac{1}{2} < b < \frac{3}{4}$; in particular $b > 0$.

Imaginary root. Put $z = \mathrm{i}y$ and write $Y = y^2 \geq 0$. The equation becomes $-Y + \sqrt{Y + b^2} = \frac{1}{2}$, that is

$$\sqrt{Y + b^2} = Y + \tfrac{1}{2}.$$

For $Y \geq 0$ the right-hand side is positive, so squaring gives an equivalent equation:

$$Y + b^2 = Y^2 + Y + \tfrac{1}{4} \quad \Longleftrightarrow \quad Y^2 = b^2 - \tfrac{1}{4}.$$

Since $b > \frac{1}{2}$, $b^2 - \frac{1}{4} > 0$, so $Y = \sqrt{b^2 - \frac{1}{4}}$ is a genuine positive value, and

$$z = \pm \mathrm{i} \left(b^2 - \tfrac{1}{4}\right)^{1/4}$$

are purely imaginary roots.

Real root. Put $z = x$ and look for a root with $x \geq -b$, where $|x + b| = x + b$. The equation becomes

$$x^2 + x + b - \tfrac{1}{2} = 0,$$

with discriminant $1 - 4(b - \frac{1}{2}) = 3 - 4b$, which is positive because $b < \frac{3}{4}$. Take the larger root

$$x_+ = \frac{-1 + \sqrt{3 - 4b}}{2}.$$

We must check that it lies in the range $x \geq -b$. Now $x_+ \geq -b$ is equivalent to $\sqrt{3 - 4b} \geq 1 - 2b$, and this is certainly true because $b > \frac{1}{2}$ makes the right-hand side negative while the left-hand side is positive. So x_+ is a purely real root.

Hence for $\frac{1}{2} < b < \frac{3}{4}$ the equation has at least one purely imaginary root and at least one purely real root, as required. (Each condition on b was used exactly once: $b > \frac{1}{2}$ for the imaginary root and for $x_+ \geq -b$, and $b < \frac{3}{4}$ for the real discriminant.)

Part (ii)

Let $z = x + \mathrm{i}y$. Then

$$z^3 = x^3 - 3xy^2 + \left(3x^2y - y^3\right)\mathrm{i}, \qquad |z+2|^2 = (x+2)^2 + y^2.$$

The second is real, so the imaginary part of the equation is

$$y\left(3x^2 - y^2\right) = 0,$$

giving $y = 0$ or $y^2 = 3x^2$. (Geometrically, z lies on the real axis or on one of the lines $y = \pm\sqrt{3}\,x$: exactly the directions whose argument is a multiple of $\frac{\pi}{3}$, which are the ones that make z^3 real.)

Case $y = 0$. The real part gives $x^3 + (x+2)^2 = 4$, that is

$$x^3 + x^2 + 4x = x\left(x^2 + x + 4\right) = 0.$$

The quadratic has discriminant $1 - 16 < 0$, so the only solution is $x = 0$, giving $z = 0$.

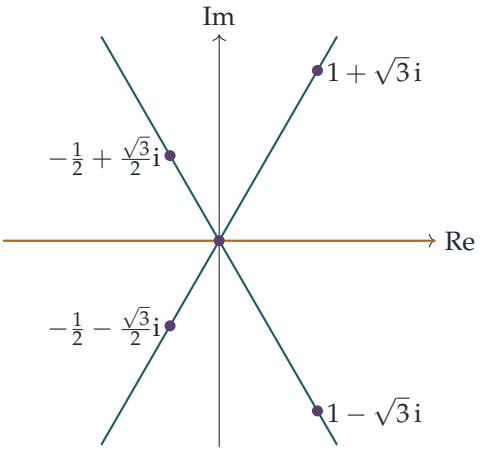
Case $y^2 = 3x^2$. Substituting $y^2 = 3x^2$ into the real part, $x^3 - 3xy^2 + (x+2)^2 + y^2 = 4$:

$$\begin{aligned} x^3 - 9x^3 + x^2 + 4x + 4 + 3x^2 &= 4 \\ -8x^3 + 4x^2 + 4x &= 0 \\ -4x\left(2x^2 - x - 1\right) &= 0 \\ -4x(2x+1)(x-1) &= 0. \end{aligned}$$

- $x = 0$ forces $y = 0$: this is $z = 0$ again.
- $x = 1$ gives $y^2 = 3$, so $z = 1 \pm \sqrt{3}\,\mathrm{i}$.
- $x = -\frac{1}{2}$ gives $y^2 = \frac{3}{4}$, so $z = -\frac{1}{2} \pm \frac{\sqrt{3}}{2}\,\mathrm{i}$.

Every step was an equivalence (we only substituted and factorised), so all of these really are solutions. The equation therefore has exactly five roots:

$$z = 0, \qquad z = 1 \pm \sqrt{3}\,\mathrm{i}, \qquad z = -\tfrac{1}{2} \pm \tfrac{\sqrt{3}}{2}\,\mathrm{i}.$$



The roots lie on the three lines $y = 0$ and $y = \pm\sqrt{3}\,x$ allowed by the imaginary part.

Checks

- In polar form, $1 + \sqrt{3}\,\mathrm{i} = 2\mathrm{e}^{\mathrm{i}\pi/3}$, so $z^3 = 8\mathrm{e}^{\mathrm{i}\pi} = -8$, and $|3 + \sqrt{3}\,\mathrm{i}|^2 = 9 + 3 = 12$; indeed $-8 + 12 = 4$.
- $-\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i} = \mathrm{e}^{2\pi\mathrm{i}/3}$ is a cube root of unity, so $z^3 = 1$, and $|\frac{3}{2} + \frac{\sqrt{3}}{2}\mathrm{i}|^2 = \frac{9}{4} + \frac{3}{4} = 3$; indeed $1 + 3 = 4$.
- In (d), try $b = \frac{5}{8}$: then $Y = \sqrt{\frac{25}{64} - \frac{16}{64}} = \frac{3}{8}$, and $-\frac{3}{8} + \sqrt{\frac{3}{8} + \frac{25}{64}} = -\frac{3}{8} + \frac{7}{8} = \frac{1}{2}$. The real root is $x_+ = \frac{-1+\sqrt{1/2}}{2} \approx -0.146$, which is above $-\frac{5}{8}$ as needed.
- A computer search for roots of the part (ii) equation over the whole region $|x|, |y| \le 3$ found exactly the five roots above.

Common slips

- Treating $|z + b|$ as if it had an imaginary part, or expanding $|z + 2|^2$ as $(z + 2)^2$. The modulus squared is $(x + 2)^2 + y^2$, a real number.
- Forgetting to check that a root of a case equation lies in the range where that case applies, as in (c) and (d).
- In (ii), dividing the imaginary part by y and so losing the real root $z = 0$.

Beyond STEP

Equations mixing z with $|z|$ (or with $\bar{z}$) are not polynomial in z, so the fundamental theorem of algebra says nothing about how many roots they have: here a "cubic" has five. The right way to think of them is as a pair of real equations in two real unknowns, and you meet this viewpoint again in first-year complex analysis, where a function is differentiable in the complex sense only when it depends on z alone and not on $\bar{z}$ (the Cauchy–Riemann equations). Needham's *Visual Complex Analysis* makes this geometric picture vivid.

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